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Clinical paper

Swedish emergency medical dispatch centres' ability to answer emergency medical calls and dispatch an ambulance in response to out-of-hospital cardiac arrest calls in accordance with the American Heart Association performance goals: An observational study

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Abstract

Aim: To investigate the ability of Swedish Emergency Medical Dispatch Centres (EMDCs) to answer medical emergency calls and dispatch an ambulance for out-of-hospital cardiac arrest (OHCA) in accordance with the American Heart Association (AHA) performance goals in a 1-step (call connected directly to the EMDC) and a 2-step (call transferred to regional EMDC) procedure over 10 years, and to assess whether delays may be associated with 30-day survival.

Method: Observational data from the Swedish Registry for Cardiopulmonary Resuscitation and EMDC.

Results: A total of 9,174,940 medical calls were answered (1-step). The median answer delay was 7.3 s (interquartile range [IQR], 3.6–14.5 s). Furthermore, 594,008 calls (6.1%) were transferred in a 2-step procedure, with a median answer delay of 39 s (IQR, 30–53 s). A total of 45,367 cases (0.5%, 1-step) were registered as OHCA, with a median answer delay of 7.2 s (IQR, 3.6–14.1 s) (AHA high-performance goal, 10 s). For 1-step procedure, no difference in 30-day survival was found regarding answer delay. For OHCA (1-step), an ambulance was dispatched after a median of 111.9 s (IQR, 81.7–159.9 s). Thirty-day survival was 10.8% (n = 664) when an ambulance was dispatched within 70 s (AHA high-performance) versus 9.3% (n = 2174) > 100 s (AHA acceptable) (p = 0.0013). Outcome data in the 2-step procedure was unobtainable.

Conclusion: The majority of calls were answered within the AHA performance goals. When an ambulance was dispatched within the AHA high-performance standard in response to OHCA calls, survival was higher compared with calls when dispatch was delayed.

Keywords: Out-of-hospital cardiac arrest (OHCA), Emergency calls, Emergency medical dispatch centre (EMDC), Dispatcher, Cardiopulmonary resuscitation, Emergency medical services, Ambulance

Introduction

The ability to rapidly reach help through an emergency medical dispatch centre (EMDC) in time-critical medical emergencies is essential. Delays in answering medical emergency calls and/or delays in the dispatch of an ambulance may have negative effects on patient outcome, especially for out-of-hospital cardiac arrest (OHCA). In Sweden, the incidence of OHCA was 56 per 100,000 population in 2021, and the overall 30-day survival rate was 10.8%.[1] Early cardiopulmonary resuscitation (CPR) and use of an automated external defibrillator by lay responders is associated with higher survival.[1–5] Thirty-day survival after a witnessed OHCA decreases as the ambulance response time increases.[6–8] Every minute of delay from the

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patient collapse to start of CPR and defibrillation reduces the chance of survival by approximately 10%.[9] Thus, time to treatment is critical in cases of OHCA.

The EUROCALL study compared time delays in two different dispatch systems in different European regions: (1) when medical emergency calls were connected directly to the EMDC (1-step procedure) and (2) when medical emergency calls were first answered by a public safety answering point (PSAP) and then transferred to the EMDC (a 2-step procedure).[10] The results showed that the median time to reach the EMDC in a 1-step procedure was 11.7 s versus 33.2 s in a 2-step procedure.[10] Thus, it seems important to assess how EMDC answer systems are organized.

According to the American Heart Association (AHA) goals, all OHCA calls should be answered within 10 s in a 1-step procedure in a high performing system, and the call answer delay should not exceed the minimal acceptable standard of 20 s. The corresponding performance goals in a 2-step procedure are 25 s and 50 s, respectively.[11] However, there is limited knowledge about whether delays in call response to the EMDC and time to ambulance dispatch affect survival in cases of OHCA.

The aim of this study was to investigate the ability of the Swedish EMDC to answer medical emergency calls and dispatch an ambulance for OHCA in accordance with the AHA performance goals in a 1-step (call connected directly to the EMDC) and a 2-step (call transferred to a regional EMDC) procedure over 10 years, and to assess whether delays are associated with 30-day survival. We hypothesise that delays in call response to the EMDC and time to ambulance dispatch affect 30-day survival.

Methods

Study design

This is an observational retrospective register study based on data over a 10-year period (2012–2021) describing medical emergency call and OHCA call handling times in Swedish EMDCs. Ethical approval was received from the Swedish ethical review authority (Dnr: 2022-02062-02).

Setting of the Swedish emergency medical dispatch centre and population

In Sweden, the national emergency number, 112, is operated and owned by the public company SOS Alarm AB (SOS Alarm) and Sweden's municipalities and regions. SOS Alarm answers all 112-emergency calls, approximately 4 million calls annually. When calling the emergency number, the call is connected to one of 15 EMDCs. Since late 2015, 112 calls are initially directed to the nearest EMDC. However, calls may be redirected nationally if not answered within a set timeframe and can be answered by any of the 15 EMDC. In most Swedish regions, the medical dispatcher at SOS Alarm handles emergency calls from start to finish (1-step procedure). In three regions, SOS Alarm acts as a PSAP and all medical calls are transferred to a regional EMDC, run by regional health organizations (2-step procedure). The emergency call handling chain in Sweden is illustrated in Fig. 1.

The 1-step procedure

In the 1-step procedure, all emergency calls are answered directly by a medical dispatcher at the EMDC, and another ambulance dispatcher is responsible for the dispatch of ambulance units. If necessary, the medical dispatcher can consult a registered nurse (RN) dispatcher anytime during the call. During the latter part of the 10-year study period, some regions have employed regional RN dispatchers to process medical emergency calls in their region and determine the medical priority for an ambulance. If the medical emergency call is determined to require ambulance assistance, a dispatch level of priority 1 to 3 is chosen depending on the urgency. In the case of a suspected OHCA, an additional ambulance resource is dispatched.

Currently, a trainee undergoes 13 weeks of education to become a medical dispatcher in the 1-step procedure. A re-certification test is performed annually. The dispatchers are supported in their medical decision-making by a criteria-based medical index.[12] The index guides dispatchers based on the symptoms described and provides direction and assistance in determining a suitable priority.[13]

The 2-step procedure

In three regions, since May 2015, a medical emergency call is transferred by the medical dispatcher at the public safety answering point (SOS Alarm) to a regional EMDC run by regional health organizations. At these regional EMDCs, all calls are answered and processed by an RN dispatcher. Ambulances are then dispatched by the same regional EMDC.

Inclusion and exclusion criteria

All medical emergency 112 calls from 1 January 2012 to 31 December 2021 were included. Exclusion criteria were as follows: medical emergency calls that did not need an ambulance response; calls transferred to the police; calls transferred to other agencies; calls categorized as solely fire/rescue; calls categorized as not an emergency call. For OHCA calls, data on call answer delay for calls processed in the 2-step procedure were not obtainable. Case numbers from the SRCR with no corresponding case number in the SOS Alarm database were excluded as missing data. OHCA witnessed by ambulance personnel was not included when presenting ambulance dispatch times or when calculating survival.

Data collection sources

Data on medical emergency calls resulting in an ambulance response were collected from the SOS Alarm database. Timestamps on, for example, the time the call reached the dispatch centre, the time when the dispatcher answered the emergency call, the time when the dispatcher transferred the medical emergency call and the time to ambulance dispatch were collected. All emergency calls have a unique case number which enabled identification of OHCA calls in the Swedish Registry for Cardiopulmonary Resuscitation (SRCR).

From the SRCR, we obtained data on 30-day survival, age, gender, and location of the OHCA. A request for data on OHCA calls processed in the 2-step procedure was declined by the regional EMDC.

SOS Alarm database

All events, indexes, time delays and the audio recordings from the emergency medical calls are available for at least 10 years, because they are classified as a medical record and are saved in the SOS Alarm database. All actions executed during an emergency call are logged with a description and a timestamp in the emergency call case folder.

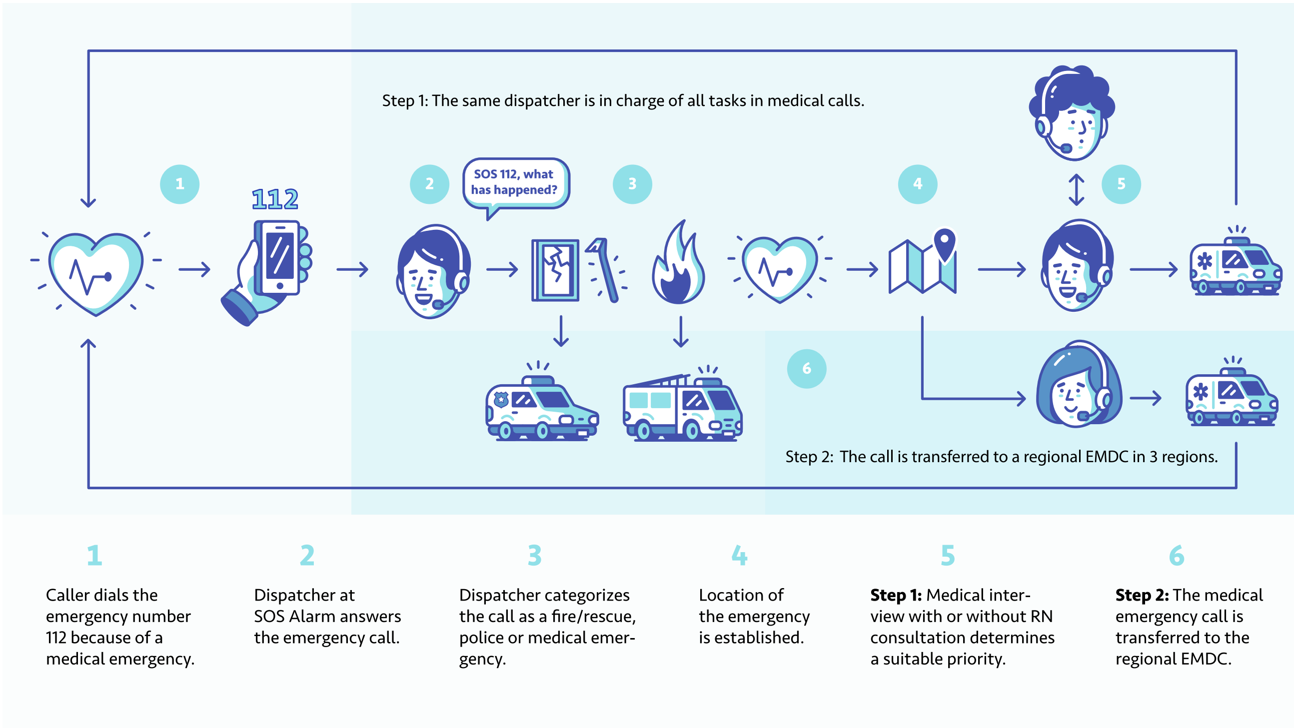

Fig. 1 – The emergency call processing chain in Sweden. EMDC, emergency medical dispatch centre; RN, registered nurse dispatcher.

Swedish register for Cardiopulmonary resuscitation

The SRCR is a national quality register. All ambulance organizations in Sweden report attempted resuscitations to the SRCR. OHCA is reported to the registry when lay responders or ambulance personnel have initiated resuscitation efforts with CPR and/or defibrillation. The SRCR follows a modified Utstein template for OHCA registration.[14] An extensive description of the SRCR has been reported previously.[15,16]

Outcome measures

The outcome measures were as follows: the proportion (%) of medical emergency calls in 1-step and 2-step procedures answered within the AHA performance goals for OHCA call handling during a 10-year period; the median time to ambulance dispatch in OHCA calls in a 1-step procedure (min:s); the median call answer delay for medical emergency calls in a 1-step versus a 2-step procedure (min:s); the median call answer delay in medical emergency calls during the day, on weekdays, months and years in a 1-step and a 2-step procedure (min:s); the unadjusted association between medical emergency call answer delay by EMDCs and 30-day survival in OHCA in a 1-step procedure; the unadjusted association between dispatch of an ambulance and 30-day survival in cases of OHCA in a 1-step procedure.

Statistical analysis

Descriptive statistics are presented as proportions (%), means (standard deviation) or medians (interquartile range). The Jonckheere-Terpstra test was used to analyse trends in changes over the years, months, days, and hours. In the 30-day survival analysis, Pearson's χ^2 test was used to compare differences in survival in groups according to the AHA performance goals. Analyses were performed using R version 4.2.2.

Results

Between 1 January 2012 and 31 December 2021, 31,150,846 Swedish 112 emergency calls were answered, of which 9,768,948 (31.4%) were medical emergency calls that later required an ambulance response or to be transferred in the 2-step procedure. The number of medical emergency calls requiring an ambulance response increased by approximately 50% over the study period, from 691,411 calls in 2012 to 1,086,416 calls in 2021.

A total of 9,174,940 (93.9%) medical emergency calls were handled in a 1-step procedure, and from May 2015 to December 2021, 594,008 (6.1%) of all the medical emergency calls were transferred in a 2-step procedure (Fig. 2). The different categories of emergency 112 calls and the number of calls for each category and year are shown in Fig. 2 and Table 1.

Time delay in medical emergency calls

The 1-step system

The centralized 1-step system answered 63.5% (n = 5,827,456) of the medical emergency calls within the AHA high performance goal (<10 s), 19.4% (n = 1,782,609) within 10–20 s (AHA minimal acceptable standard) and 17.1% (n = 1,564,676) > 20 s. The median delay was 7.3 s (interquartile range [IQR], 3.6–14.5 s). Variation in answering times was observed with regard to the following: 2014 and 2021, during the summer months, during weekends and in the late and early hours of the day. There were significant changes over the

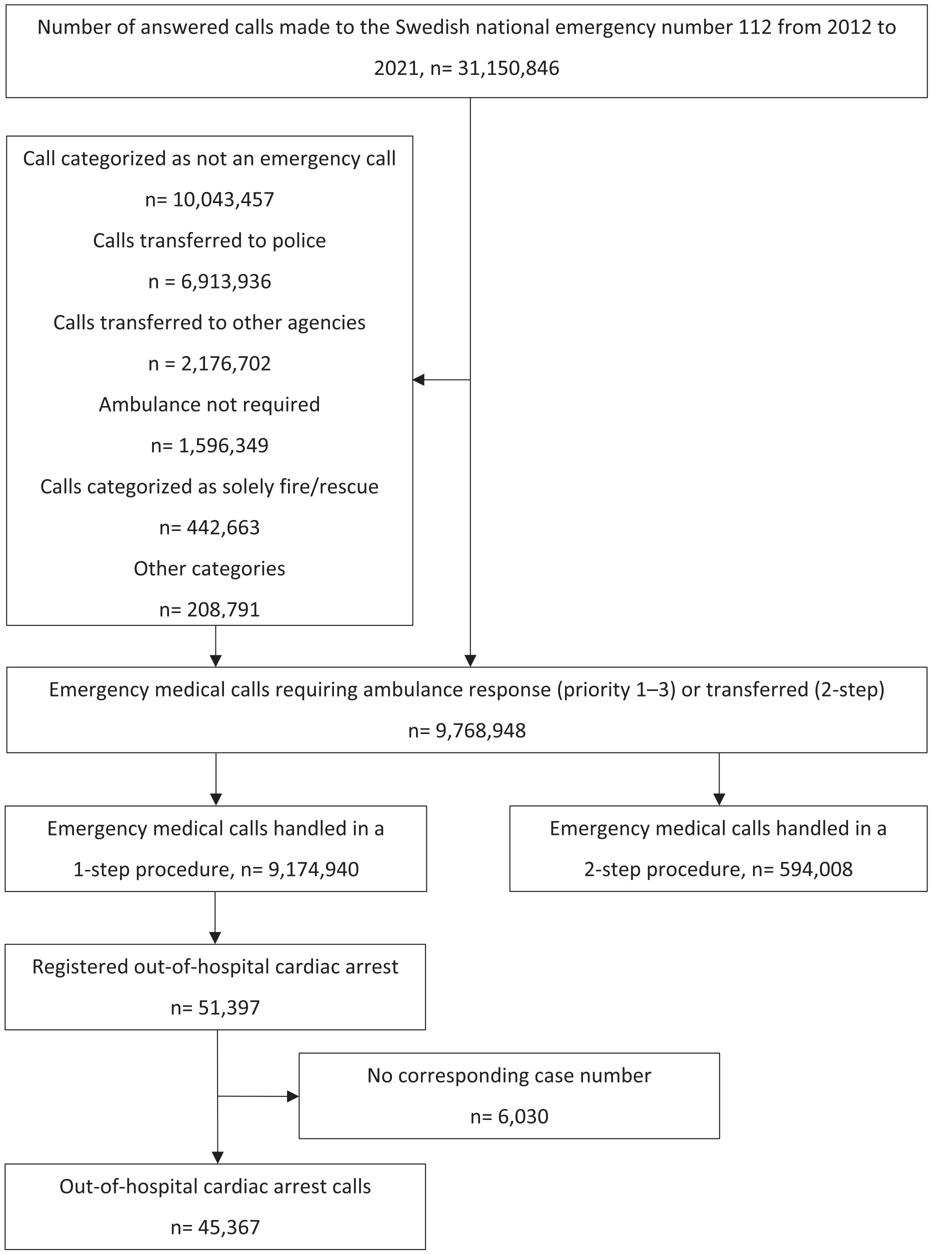

Fig. 2 – Flowchart of the study.

10-year study period on the different time distributions (Fig. 3). The yearly variation over time is presented in Table 1.

The 2-step system

The median delay time for the PSAP to transfer the medical emergency call (2-step procedure) was 25 s; in addition, it took 4 s (IQR, 3.6–5.3 s) for the regional EMDC to answer the call. Overall, the regional EMDC answered 9.6% (*n* = 56,898) of the medical emergency calls within the AHA high performance goal (25 s for the 2-step procedure), 61.8% (*n* = 367,344) within 25–50 s (AHA minimal acceptable standard) and 28.4% (*n* = 168,955) > 50 s. The median total time until the regional EMDC answered the call was 39 s (IQR, 30–53 s). The distribution of medical emergency calls answered within AHA time goals, median and mean answering times for each year are presented in Table 1. Overall, the same variation in answering times for years, months, weekdays, and time of day as in the 1-step procedure is observed for the 2-step procedure, and there were also significant changes over the 10-year study period on the different time distributions (Fig. 3).

Time delay in OHCA calls

The 1-step system

Of all medical emergency calls, 51,397 were later registered in the SRCR as OHCA. After excluding registered OHCA cases with no corresponding case number in the SOS Alarm database (*n* = 6,030 or 11.7%), 45,367 (88.3%) OHCA calls were included (Fig. 2). For OHCA calls, 64.4% (*n* = 29,194) were answered within the AHA high performance goal (<10 s), 19.3% (*n* = 8778) within 10–20 s (AHA

Table 1 – Emergency 112-call answering delays for medical and OHCA calls in a 1-step procedure and medical calls in a 2-step procedure.

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	
Medical calls in 1-step procedure											**P for trend**
Call answer delay											
Number of calls	691,411	748,017	839,394	833,427	947,255	994,630	999,254	1,018,146	1,016,990	1,086,416	
0–10 s, %	76.5	64.1	59.8	64.4	54.2	59.1	63.1	72.3	69.7	55.6	
10–20 s, %	15.7	18.6	16.9	16.9	24.6	23.8	20.5	18.0	18.3	19.1	
>20 s, %	7.9	17.2	23.3	18.7	21.2	17.1	16.3	9.6	12.1	25.3	
Mean, s (SD)	8.2 (11)	12.3 (17)	16.3 (26)	14.1 (22)	15.1 (18)	13.0 (14)	12.0 (15)	9.0 (11)	9.7 (11)	16.7 (24)	
Median, s (IQR)	5.2 (3–10)	6.8 (3–14)	7.2 (3–18)	6.5 (3–15)	9.2 (7–17)	8.5 (6–15)	7.3 (4–14)	5.9 (3–11)	5.7 (3–12)	8.2 (4–20)	<0.0001
OHCA calls in 1-step procedure											
Call answer delay											
Number of calls	3758	4012	4634	4673	4525	4867	5047	4697	4715	4439	
0–10 s, %	76.7	65.7	60.4	64.0	56.0	59.8	63.1	72.3	70.4	57.1	
10–20 s, %	15.8	18.3	17.0	17.3	24.2	23.7	20.4	17.6	18.5	19.6	
>20 s, %	7.5	15.9	22.6	18.7	19.8	16.5	16.4	10.1	11.1	23.3	
Mean, s (SD)	8.1 (9)	11.6 (15)	15.7 (23)	14.7 (24)	14.6 (15)	12.8 (13)	12.1 (14)	9.1 (9)	9.4 (11)	15.2 (20)	
Median, s (IQR)	5.2 (3–10)	6.4 (3–14)	7.0 (3–18)	6.7 (3–15)	8.9 (7–17)	8.3 (6–15)	7.3 (4–14)	5.8 (3–11)	5.6 (3–12)	7.7 (3–19)	0.0058
Ambulance dispatch time (1-step)											
<70 s, %	19.7	18.2	15.4	13.3	13.1	16.9	19.1	13.9	8.7	7.7	
70–100 s, %	23.4	25.5	23.9	23.3	24.2	26.3	25.1	24.9	25.2	21.5	
>100 s, %	56.9	56.2	60.5	63.2	62.5	56.7	55.7	61.2	66.0	70.8	
Mean, s (SD)	156.9 (274)	144.2 (165)	151.0 (164)	156.0 (195)	154.0 (184)	153.6 (207)	159.3 (294)	161.4 (256)	182.0 (444)	179.9 (276)	
Median, s (IQR)	110.4 (77–166)	108.3 (78–158)	115.6 (82–166)	117.7 (86–170)	116.5 (85–165)	108.2 (79–159)	108.8 (77–160)	115.3 (84–168)	121.1 (90–176)	126.1 (95–183)	<0.0001
Medical calls in 2-step procedure											
Call answer delay in medical calls by PSAP (2-step)											
Number of calls	NA	NA	NA	35,305	61,168	65,520	99,200	101,930	109,094	121,791	
0–10 s, %	NA	NA	NA	71.9	57.5	59.2	70.0	80.7	78.1	65.7	
10–20 s, %	NA	NA	NA	13.7	24.2	25.2	17.1	12.8	13.1	14.8	
>20 s, %	NA	NA	NA	14.4	18.3	15.6	12.9	6.5	8.8	19.4	
Mean, s (SD)	NA	NA	NA	11.6 (19)	14.0 (15)	12.7 (12)	10.4 (12)	7,3 (8)	7.9 (10)	13.5 (20)	
Median, s (IQR)	NA	NA	NA	5.2 (3–11)	8.7 (7–16)	8.5 (7–15)	6.3 (3–12)	4.9 (3–8)	4.0 (2–9)	5.5 (3–15)	<0.0001
Time for PSAP to call regional EMDC (2-Step)											
Mean, s (SD)	NA	NA	NA	32.7 (34)	31.3 (32)	30.1 (28)	29.8 (27)	32.1 (30)	32.7 (31)	31.9 (29)	
Median, s (IQR)	NA	NA	NA	23.2 (17–34)	23.4 (18–34)	23.8 (18–33)	24.1 (18–33)	25.4 (19–35)	25.5 (20–36)	25.5 (20–35)	<0.0001
Reginal EMDC call answer delay (2-step)											
0–15 s, %	NA	NA	NA	98.5	98.8	99.0	99.3	99.3	99.4	99.1	

(*continued on next page*)

Table 1 (continued)

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	
15–30 s, %	NA	NA	NA	1.3	1.0	0.8	0.6	0.6	0.5	0.7	
>30 s, %	NA	NA	NA	0.2	0.2	0.2	0.1	0.1	0.1	0.2	
Mean, s (SD)	NA	NA	NA	5.7 (3)	5.6 (3)	5.3 (3)	4.8 (2)	4.5 (2)	4.3 (2)	4.5 (3)	
Medians, s (IQR)	NA	NA	NA	5.0 (4–6)	5.0 (4–6)	4.7 (4–6)	4.3 (4–5)	4.0 (3–5)	3.8 (3–5)	3.9 (3–5)	<0.0001
Total call answer delay in 2-step (PSAP + regional EMDC)											
0–25 s, %	NA	NA	NA	13.0	5.2	4.8	10.1	11.2	12.1	9.4	
25–50 s, %	NA	NA	NA	57.0	60.9	64.8	63.6	65.4	62.5	57.5	
>50 s, %	NA	NA	NA	30.0	33.9	30.5	26.3	23.4	25.3	33.1	
Mean, s (SD)	NA	NA	NA	49.9 (40)	50.9 (36)	48.0 (31)	45.1 (30)	43.9 (32)	44.8 (33)	49.9 (36)	
Median, s (IQR)	NA	NA	NA	37.8 (29–55)	41.5 (33–57)	40.8 (33–54)	38.2 (30–51)	36.9 (29–49)	37.1 (29–50)	40.4 (31–57)	<0.0001

EMDC, Emergency Medical Dispatch Centre; IQR, interquartile range; OHCA, out-of-hospital cardiac arrest; PSAP; public safety answering point; SD, standard deviation.

minimal acceptable standard) and 16.3% (n = 7 394) > 20 s. The median delay was 7.2 s (IQR, 3.6–14.1 s). For OHCA calls, an ambulance was dispatched within the AHA high performance goal of 70 s (10 s + 60 s) in 15.4% (n = 6256), 25.4% (n = 10,346) within 70–100 s (AHA minimal acceptable standard) and 59.2% (n = 24,090) > 100 s. The median ambulance dispatch time was 111.9 s (IQR, 81.7–159.9 s). For each year, the distribution of OHCA calls answered within AHA time goals, the median and mean answering times, as well as ambulance dispatch times in the 1-step procedure are presented in Table 1 and Fig. 4.

30-day survival after OHCA

The 1-step system

No significant difference in unadjusted 30-day survival was found when OHCA calls were answered within 10 s (9.8%, AHA high performance goal) and when calls exceeded 20 s (9.5%, p = 0.531, minimal acceptable standard).

Thirty-day survival was 10.8% (n = 664) when an ambulance was dispatched within 70 s (AHA high performance goal) versus 9.3% (n = 2174) when dispatch exceeded 100 s (p = 0.0014, minimal acceptable standard). The effect of emergency call answering delay and ambulance dispatch delay, grouped in deciles, on 30-day survival is shown in Fig. 5.

Discussion

In this nationwide registry-based study, we show that over a period of 10 years, Swedish EMDCs have been able to answer most medical emergency calls in accordance with the AHA goals, but the time to dispatch an ambulance exceeded the AHA goal in a majority of OHCA calls. In addition, we show that delayed time to dispatch of an ambulance was associated with decreased 30-day survival.

In the 1-step procedure, most medical emergency calls and OHCA calls were answered in accordance with the AHA high performance goals. In the 2-step procedure, when the calls are first answered by a PSAP and then transferred to the regional EMDC, the AHA high performance goal was not achieved. In the 2-step procedure, when the medical dispatcher at the PSAP identifies a medical emergency and the need to transfer the call to the regional EMDC is a time-consuming process. The median PSAP time delay to call the regional EMDC was 25 s. The median regional EMDC call answer delay was 4 s, which in the 1-step procedure would be considered a high performing system. Similar results are reported both from parts of Europe and New York in the United States of America. [10,17] This study does not investigate possible reasons for the delay at the PSAP in the 2-step procedure. We can only conclude that the 2-step procedure prolongs the time to reach the EMDC, this result needs to be investigated further to identify reasons, and to find possible improvement measures. In the EUROCALL study, possible reasons for delayed response time in the 2-step procedure were short-term variation in call volume and technical or organizational differences.[10] It is not known whether the prolonged call answer delay in the 2-step procedure affects patient outcomes, but it may be relevant in cases of OHCA because the patient's outcome is critically time-dependent,[10] and survival decreases by 10% per minute delay from patient collapse to start of CPR.[9] Efforts should be made to reduce the processing time at the PSAP to reach the AHA high performance goal for the 2-step procedure.

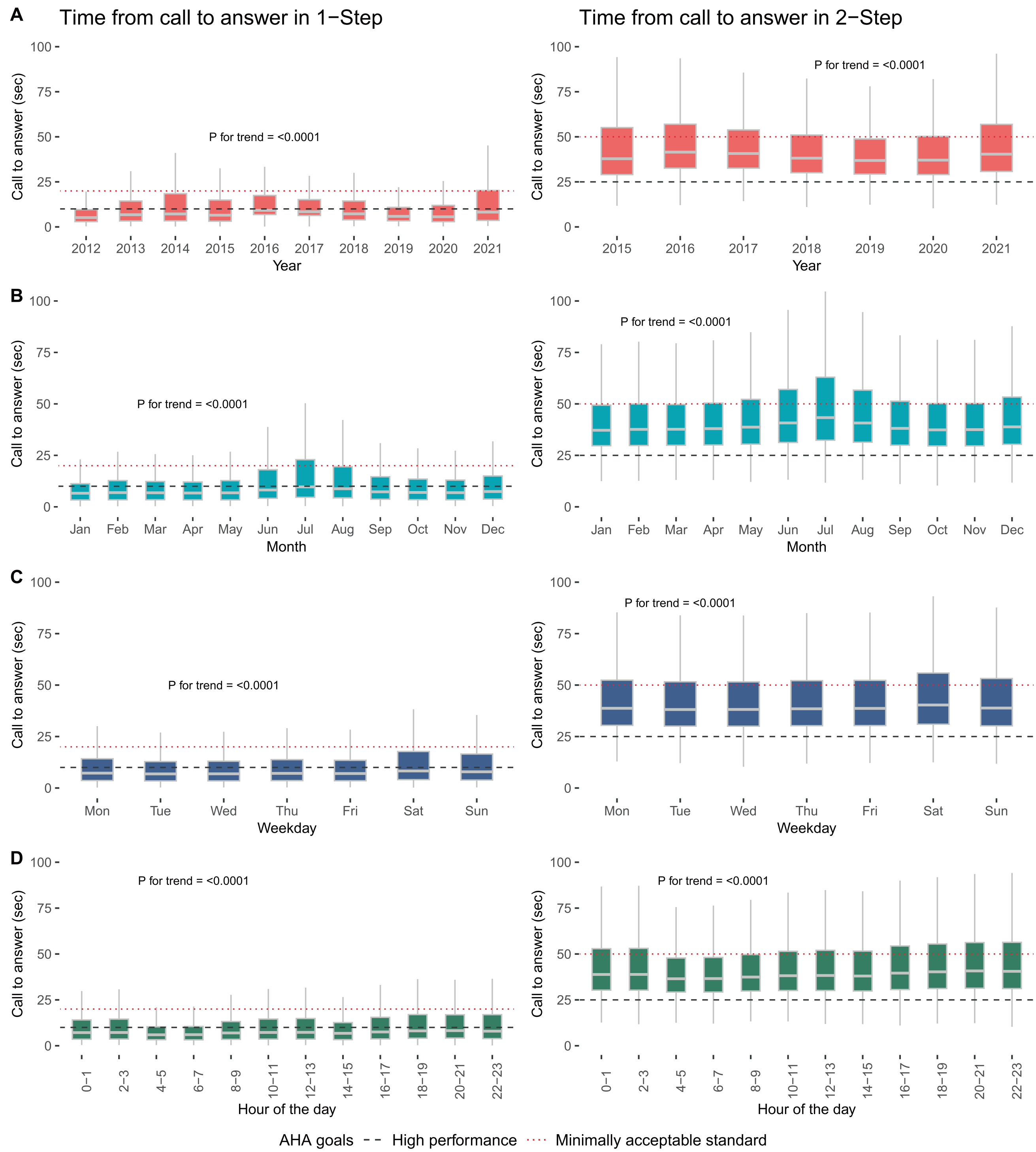

Fig. 3 – Emergency medical call answer delay in a 1-step and 2-step procedure according to year, month, weekday and time of day.

Over the 10-year study period, an increase in the medical emergency call answer delay for 1-step procedure was observed in 2014 and 2021. A possible explanation for the increase in 2014 could be that the regions that had implemented a 2-step procedure had to temporally halt their ability to answer and process medical emergency calls. The medical emergency calls were subsequently answered in a 1-step procedure until the 2-step procedure was implemented again in 2015. There are several possible explanations for the increase in the emergency call answer delay in 2021. Due to sick leave, the Covid-19 pandemic probably affected the number of medical dispatchers available to answer emergency calls. The pandemic also influenced the amount of emergency calls during 2021.

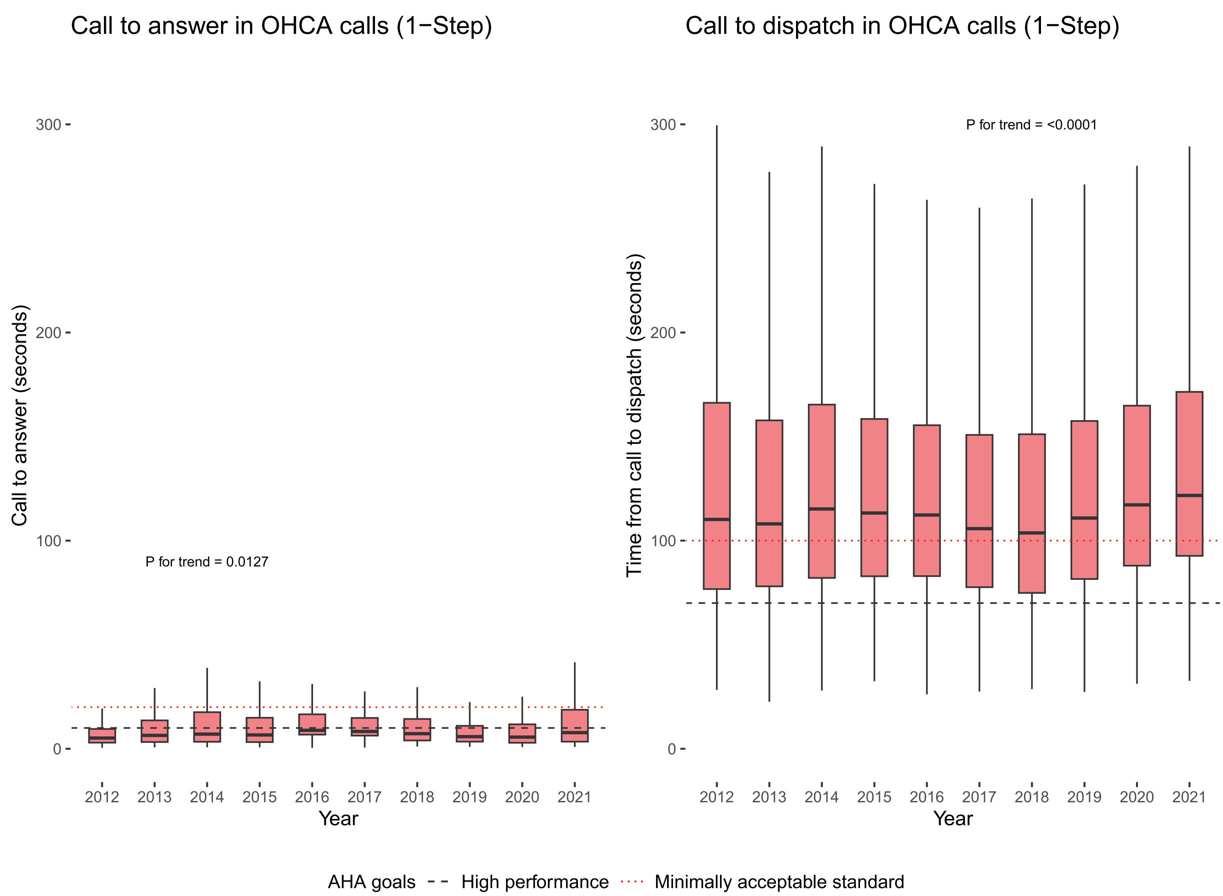

Fig. 4 – Emergency call answer delay and time to dispatch for out-of-hospital cardiac arrest calls (1-step procedure) over 10 years.

A newly developed medical decision support tool was introduced at SOS Alarm EMDC during 2021, which involved a novel way of collecting information during the medical emergency call that could have increased the dispatcher's processing time, subsequently affecting the time the dispatcher could answer subsequent emergency calls.

Our results show an increased call answer delay during early mornings, late evenings, at weekends, and in the summer months, which could also be due to a lower staffing level. Similar variability depending on the day of the week, weekends and weekdays or day calls versus night calls was not seen in the EUROCALL study, based on data from 11 countries.[10]

During the 10-year study period, a substantial increase in the medical emergency calls requiring an ambulance response was observed. We can only speculate, but an increase in emergency demands could partially give an explanation to the increase in the emergency call answer delay and subsequent dispatch delay.

A significant difference in 30-day survival was found when an ambulance was dispatched within the AHA high performance goal compared with when the AHA minimal acceptable standard was exceeded. Overall, in most OHCA calls, the time to dispatch an ambulance exceeded the AHA acceptable standard. We have also observed an increase in ambulance dispatch times in OHCA calls in recent years; the median was 108 s in 2017 and 126 s in 2021. The reason for this trend is unknown, but these results show a need for improvement. Previous studies have shown that 30-day survival after a witnessed OHCA decreases as ambulance response time increases.[7] Previous studies have also shown that shorter ambulance dispatch times is associated with higher survival rates.[18–20] A future possible solution could be an automated dispatch system whereby the ambulance is dispatched automatically when the location of the emergency is available, and the medical priority is established.

Limitations

The analysis on 30-day survival is unadjusted and therefore does not take other predictors for survival (e.g. initial rhythm or age) into consideration. We were not able to obtain data on medical emergency call answer delays and ambulance dispatch times in OHCA calls processed in the 2-step procedure. Collecting emergency call data in OHCA calls processed in the 2-step procedure required authorization from the regional EMDC, which was declined. However, there should be no difference in the answer time between the reported medical calls and OHCA calls. This study does not present results on individual OHCA patients. How extensively prolonged emergency call answer delay in OHCA calls affects individual patient outcome is unclear. The findings of this study are limited to the Swedish dispatch system and are therefore only generalizable to similar dispatch systems.

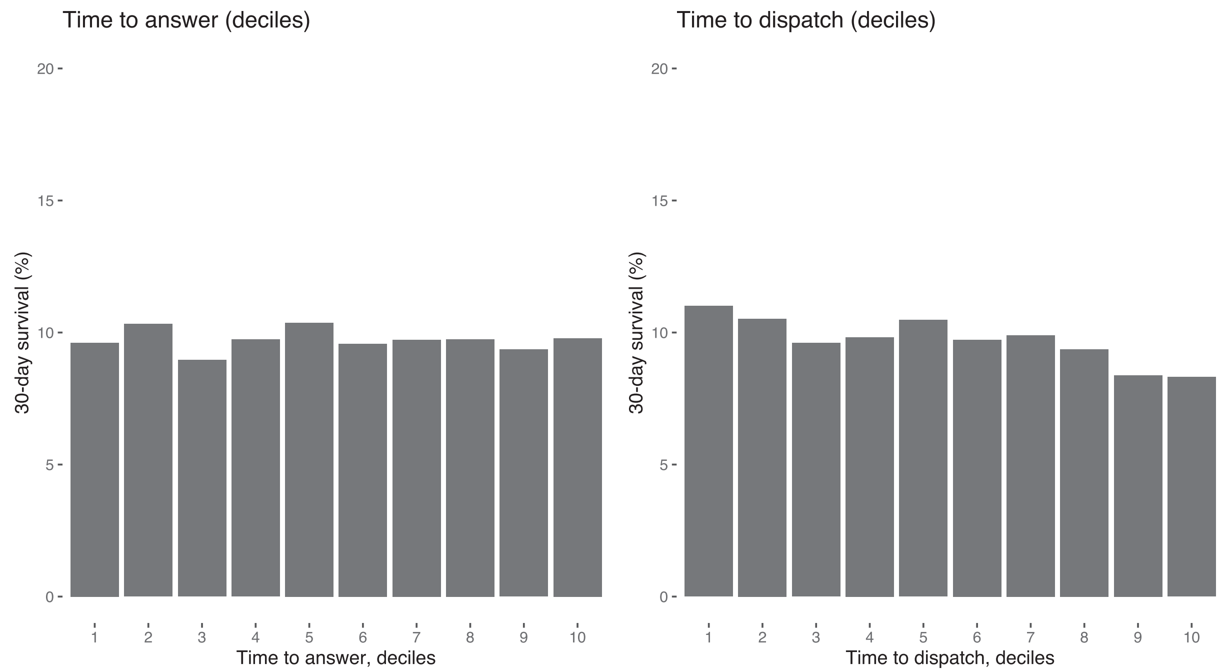

Fig. 5 – The effect of emergency call answer delay and time to dispatch in out-of-hospital cardiac arrest calls (1-step procedure) on 30-day survival. Emergency call answer delay and time to dispatch are grouped into deciles on the x-axis.

Conclusion

The majority of calls were answered within the AHA performance goals. When an ambulance was dispatched within the AHA high-performance standard in response to OHCA calls, survival was higher compared with calls when dispatch was delayed.

Author contributions

Conceptualization, FB, MJ, AC, MR, LS, JH, and AN; Methodology, FB, MJ, AC, MR, LS, and AN; Data Curation, FB; Formal Analysis, FB, AN and MJ; Writing – Original Draft, FB; Writing – Review & Editing, FB, MJ, AC, MR, LS, GR, PN, SF, JH and AN.

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CRediT authorship contribution statement

Fredrik Byrsell: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Martin Jonsson:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Andreas Claesson:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Mattias Ringh:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Leif Svensson:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Gabriel Riva:** Writing – review & editing. **Per Nordberg:** Writing – review & editing. **Sune Forsberg:** Writing – review & editing. **Jacob Hollenberg:** Writing – review & editing, Conceptualization. **Anette Nord:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: F. Byrsell is an employee at SOS Alarm AB. None of the other authors have any conflicts of interest to declare.

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